Quantum Junction Challenge: Condensed Codex Session Report

Codex run summary

Built 2026-06-06 06:33:07

Executive Summary

The session produced selected peak bitstrings for **40/49** challenge files. The remaining blanks are `64_40, 104_49, 48_42, 56_43, 64_44, 72_45, 80_46, 88_47, 96_48`. The selected set contains **10** exact statevector answers and **30** tensor-network or MPS-derived candidates. Among the approximate selected rows, **17** are from the canonical Quimb GPU pass, **11** from canonical Quimb CPU, and **1** from an RCM-order CPU fallback. Exact statevector peaks had mean peak probability 0.513 and mean gap to the second bitstring 0.459.

The main result is a usable candidate set, not a proof of correctness for every approximate row. Easy and moderate instances often had large sampled peak fractions and/or corroborating evidence. Several hard rows are weak: they were selected because they were the best available evidence, but their sampled top fractions are near one or two samples out of 512/1024. The session later pivoted away from long-running graph-MPS sweeps after the user set a practical "about 2 hours" per-attempt budget.

Candidate Selection Rule

The collector selected one answer per challenge using source priority, then evidence consistency. The priority order used in the current collector was: exact statevector, canonical Quimb GPU, optimized-U3 GPU, canonical Quimb CPU, MPO-unswap, RCM/MST/ degree/mid/fast/identity CPU fallbacks, sparse beam, and low-priority Aer MPS pilot evidence. Exact answers were accepted directly. Approximate answers were chosen as sampled top bitstrings in Qiskit/counts order, with the right-most bit corresponding to qubit 0. When multiple methods agreed on the same bitstring, the report records that as higher evidence count.

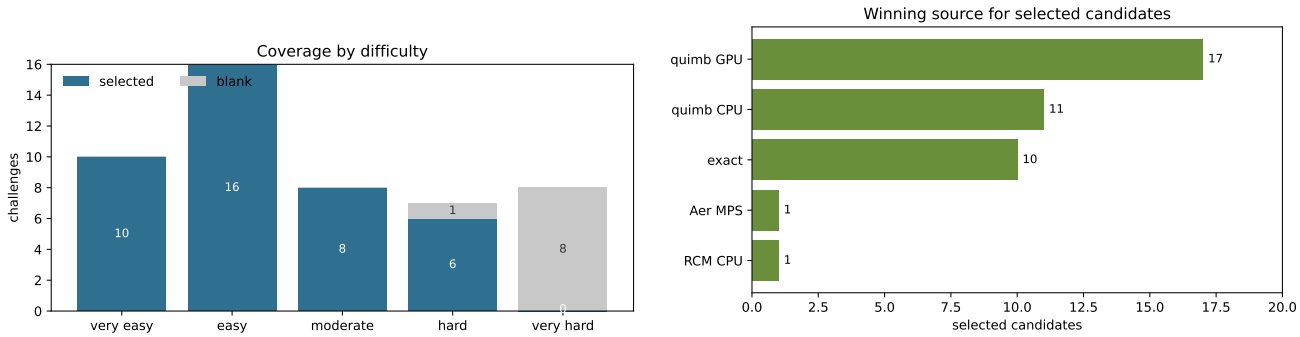


Figure 1: Coverage and winning evidence sources. The nine missing rows are all hard or very hard.

What Was Tried

Method	Purpose and implementation	Outcome

Exact statevector	Used Aer/statevector on small enough circuits to establish ground-truth peaks and bit ordering.	10 exact answers. Used as calibration and highest-priority evidence.
Canonical graph/tree MPS	Ran Quimb tree-ordered MPS sampling on CPU and GPU, using sampled top bitstrings as candidate peaks.	Main source of approximate answers; strong on many easy/moderate rows and weak but usable on some hard rows.
Ordering fallbacks	Tried RCM, MST, degree, mid, fast, and identity orderings to change contraction behavior.	Added corroboration; RCM supplied the selected 56_38 weak candidate. Long jobs were later cancelled.
Sparse computational-basis beam	Added a C++ top-K sparse branch propagator through RX/RZ/CX.	Rejected for answer selection: on known 16_28 the exact answer was only rank 3.
Static QASM forensics	Counted gates, angle classes, pair repetitions, connectivity, leading one-qubit prefixes, and operation 5-gram similarity.	Confirmed all circuits are RX/RZ/CX/SWAP style, very-hard files are a distinct dense-obfuscation regime, and Clifford shortcuts are not appropriate.
Optimized-U3 transpilation	Transpiled the remaining blanks to U3+CX to reduce single-qubit gate count.	Reduced total gate count by roughly 39–42% on blanks; not yet selected as evidence.
MPO unswapping	Patched the local Kremer-Dupuis style MPO cancellation/unswapping code: parameterized SABRE trials, fixed Quimb local expectation API, fixed quantum-op progress counting.	Calibrated on known 16_28 in 93.26 s and recovered the exact answer. Unresolved runs were still active at report build time and are not counted as solved.

Static Structure

The QASM files use only `rx`, `rz`, `cx`, and sparse `swap` gates. Very-hard instances are much larger and more internally similar to each other than to the easier sets. They also have dense leading single-qubit prefixes and high all-to-all pair coverage. This drove the pivot toward structure-aware MPO cancellation rather than only waiting for generic graph-MPS contractions.

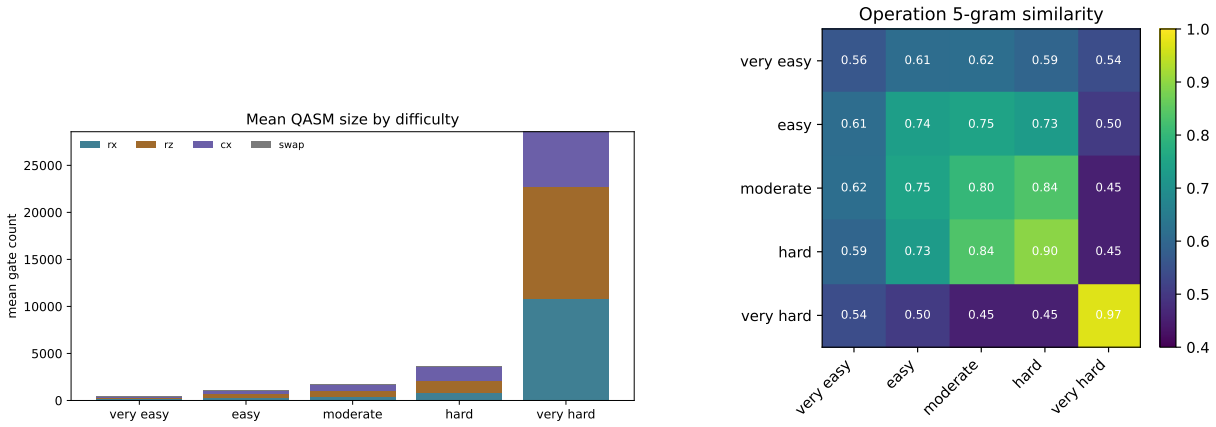


Figure 2: Static QASM summary. Very-hard files are large and operation-sequence-similar to each other.

Exact Baseline

The exact rows were used both as final answers and as calibration targets. The peak is visibly separated from the runner-up on these cases, which made them useful for validating bit order and method behavior.

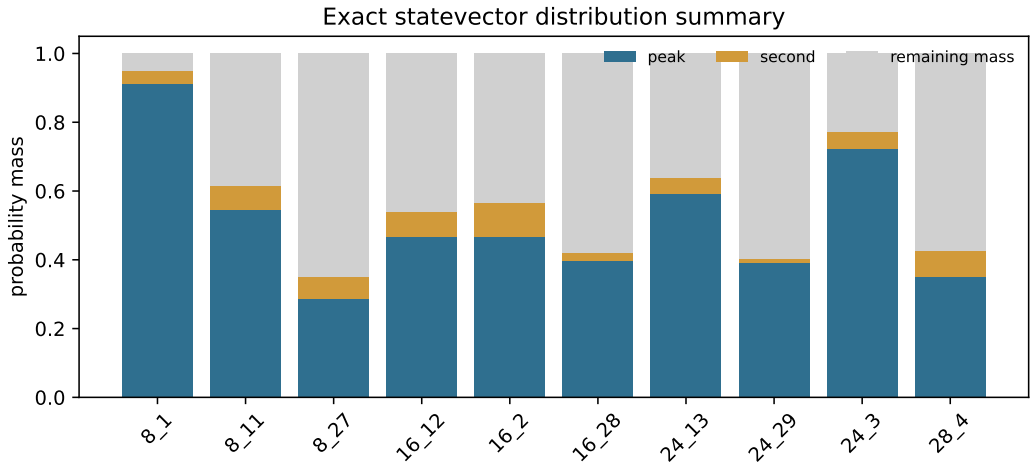


Figure 3: Exact statevector probability mass: peak, second-largest bitstring, and remaining mass.

Challenge	q	exact peak	peak prob.	second prob.	gap
8_1	8	10101101	0.9126	0.0379	0.8747
8_11	8	01001110	0.5454	0.0687	0.4768
8_27	8	11001001	0.2879	0.0627	0.2252
16_12	16	1111000101101011	0.4665	0.0730	0.3936
16_2	16	1010101011001000	0.4670	0.0976	0.3694
16_28	16	1101001111011100	0.3962	0.0231	0.3731
24_13	24	111110011111001011010001	0.5912	0.0472	0.5441
24_29	24	110100010111100001001001	0.3901	0.0134	0.3767
24_3	24	011110010000101010001000	0.7240	0.0471	0.6769
28_4	28	1111111000101010110110011111	0.3511	0.0735	0.2776

Approximate Tensor-Network Results

For approximate rows, the sampled top fraction was the main confidence signal. Strong rows have a clear heavy bitstring. Weak hard rows are retained because they are currently the best available candidates, but they should be interpreted as low-confidence until corroborated.

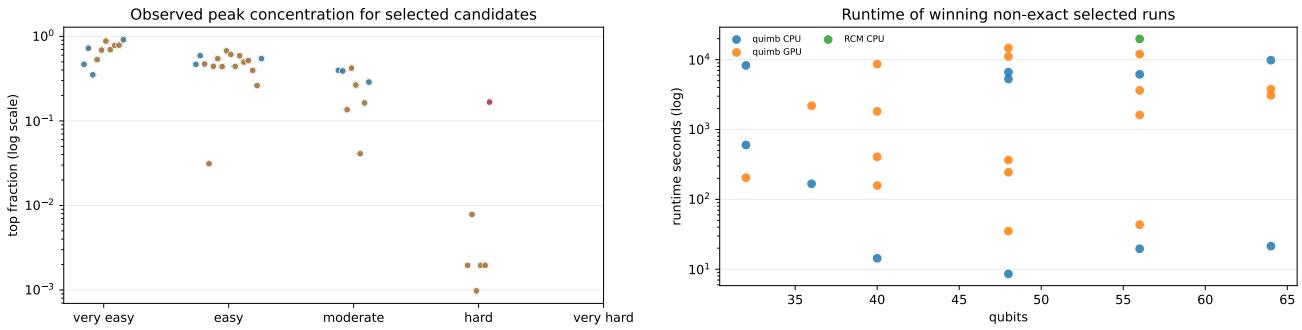


Figure 4: Peak concentration and runtime for selected rows. The hard set contains several low-concentration candidates.

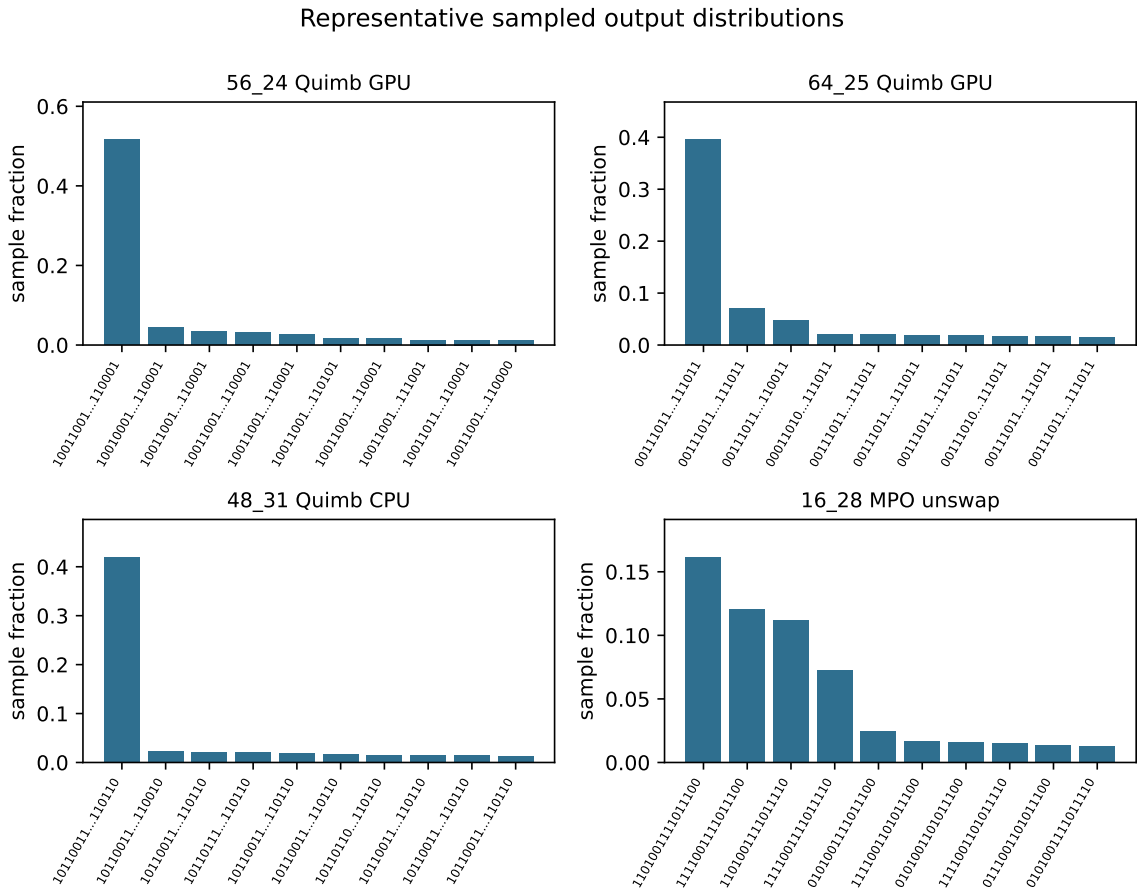


Figure 5: Representative sampled distributions. The first three are selected Quimb candidates; the fourth is the MPO-unswap validation on 16_28.

Weak Selected Rows

Challenge	difficulty	source	top frac.	evidence	candidate
40_35	hard	quimb GPU	0.001953	3	0101100110101111111000010101001110001001
48_36	hard	quimb GPU	0.007812	3	001111011011001100110011001000101001101010111000
48_37	hard	quimb GPU	0.0009766	3	100101001010001101010100100101001001000101010001
56_38	hard	RCM CPU	0.001953	1	10011001011110000000001001101010110000100100111010110110001
56_39	hard	quimb GPU	0.001953	3	10010111001101010101111111010110110110110000101001010000
64_41	hard	Aer MPS	0.167	1	0001000111010011100110010011110111100000010001101000110101010110010
32_5	very easy	quimb CPU	0.5322	1	00111000101010100001000000010000
36_6	very easy	quimb CPU	0.6885	1	100110100111101001001101110011001000
40_7	very easy	quimb CPU	0.8789	1	0110111011010001010011111110010011000111
48_8	very easy	quimb CPU	0.6973	1	000100100110111001001111111100101011001110010100
56_9	very easy	quimb CPU	0.7812	1	10010110101100101110100110010110011101100110001101010100
64_10	very easy	quimb CPU	0.7871	1	0011010010110001110010111001100100101100010111110110010100101011

Rejected or Not-Yet-Selected Paths

The sparse beam search was attractive because it was fast, but its pruning bias failed calibration. In known 16_28, the exact answer was rank 3 rather than rank 1, so no sparse-beam guesses were promoted to selected answers. U3 optimization reduced the remaining blank gate counts substantially, but its GPU pass was not completed before this report snapshot.

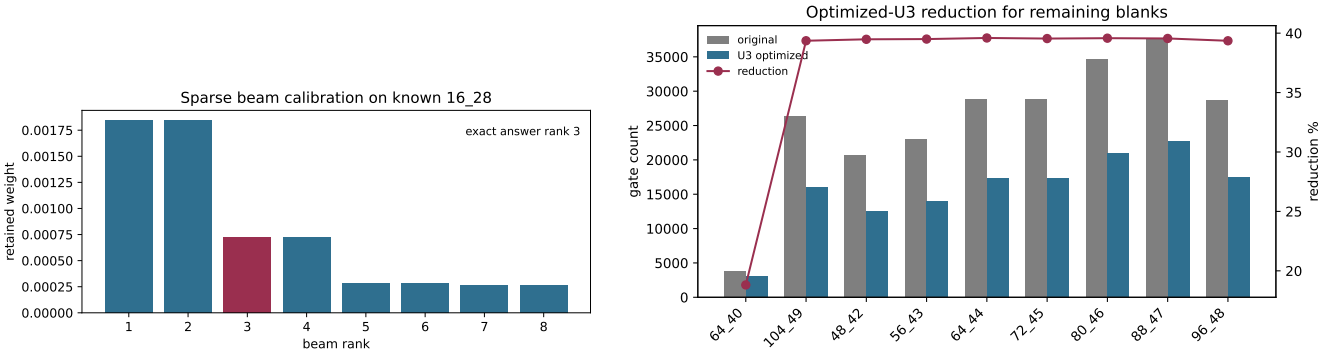


Figure 6: Left: sparse beam failed calibration. Right: U3 optimization reduced the remaining blank circuits but was not yet selected as evidence.

Blank	q	original gates	U3 gates	reduction
64_40	64	3758	3051	18.8%
104_49	104	26361	15985	39.4%
48_42	48	20733	12547	39.5%
56_43	56	23004	13917	39.5%
64_44	64	28787	17389	39.6%
72_45	72	28777	17397	39.5%
80_46	80	34586	20898	39.6%

88_47	88	37660	22764	39.6%
96_48	96	28703	17406	39.4%

Current Slurm Snapshot

The report is an as-of snapshot. Any active unresolved MPO-unswap jobs below were still running or pending when the PDF was built, and their outputs are not counted in the **40/49** result.

JobID	state	elapsed	CPUs	name	node/reason
34614255	PENDING	0:00	2	peaked_unswap_rollup	(Dependency)
34614254_[46-48%5]	PENDING	0:00	16	peaked_unswap_unresolved	(JobArrayTaskLimit)
34614254_45	RUNNING	3:12	16	peaked_unswap_unresolved	erc-hpc-comp202
34614254_21	RUNNING	4:44	16	peaked_unswap_unresolved	erc-hpc-comp192
34614254_41	RUNNING	4:44	16	peaked_unswap_unresolved	erc-hpc-comp195
34614254_43	RUNNING	4:44	16	peaked_unswap_unresolved	erc-hpc-vm064
34614254_44	RUNNING	4:44	16	peaked_unswap_unresolved	erc-hpc-vm065

Method Summary

Source	selected winners	candidate evidence rows	attempted evidence rows
Aer MPS	1	6	6
exact	10	10	10
MPO unswap	0	1	1
quimb CPU	11	36	36
degree CPU	0	1	1
fast CPU	0	5	5
quimb GPU	17	21	21
RCM CPU	1	16	16
sparse beam	0	5	5

Selected Candidate Table

Full candidate bitstrings are shown in wrapped monospace chunks. Blank rows remain unsolved.

Challenge	difficulty	q	candidate	source	validation	top frac.	evidence
8_1	very easy	8	10101101	exact	exact	0.9126	2
16_2	very easy	16	1010101011001000	exact	exact	0.467	2
24_3	very easy	24	011110010000101010001000	exact	exact	0.724	2
28_4	very easy	28	1111111000101010110110011111	exact	exact	0.3511	2
32_5	very easy	32	00111000101010100001000000010000	quimb CPU	unknown	0.5322	1
36_6	very easy	36	100110100111101001001101110011001000	quimb CPU	unknown	0.6885	1
40_7	very easy	40	0110111011010001010011111110010011000111	quimb CPU	unknown	0.8789	1
48_8	very easy	48	000100100110111001001111111100101011001110010100	quimb CPU	unknown	0.6973	1
56_9	very easy	56	10010110101100101110100110010110011101100110001101010100	quimb CPU	unknown	0.7812	1
64_10	very easy	64	0011010010110001110010111001100100101100010111110110010100101011	quimb CPU	unknown	0.7871	1
8_11	easy	8	01001110	exact	exact	0.5454	4
16_12	easy	16	1111000101101011	exact	exact	0.4665	4
24_13	easy	24	111110011111001011010001	exact	exact	0.5912	3
32_14	easy	32	00000101001101000001011111101100	quimb GPU	unknown	0.4717	3
36_15	easy	36	110110011011111111101100101111010111 1	quimb GPU	unknown	0.03125	3
40_16	easy	40	0101110101001110011000111011100110010110	quimb GPU	unknown	0.4424	4

40_17	easy	40	0010010101010111001001000010001000001001	quimb GPU	unknown	0.5459	4
40_18	easy	40	0100000110010010001101111000111111001110	quimb GPU	unknown	0.4395	3
48_19	easy	48	011001010111101100111110000001011101001110010000	quimb GPU	unknown	0.6738	2
48_20	easy	48	101010100101001010000110101000001011110010000000	quimb GPU	unknown	0.6123	3
48_21	easy	48	111010101110101011110101000001101000100000001001	quimb GPU	unknown	0.4414	3
56_22	easy	56	11100010010011001001111001011011001100000001000101 11111011	quimb GPU	unknown	0.5898	2
56_23	easy	56	01001101001111111100000101001110111011100011111100001101	quimb GPU	unknown	0.4971	2
56_24	easy	56	10011001001111101111110111010111011010100110010 11110001	quimb GPU	unknown	0.5176	4
64_25	easy	64	0011101111110000110110111101011000010000001010011 0111000001111011	quimb GPU	unknown	0.3965	4
64_26	easy	64	0110101010100011010111011000011100010110110110011 1100011001100110	quimb GPU	unknown	0.2617	3
8_27	moderate	8	11001001	exact	exact	0.2879	2
16_28	moderate	16	1101001111011100	exact	exact	0.3962	6
24_29	moderate	24	110100010111100001001001	exact	exact	0.3901	4
32_30	moderate	32	10011000010011110111101111010111	quimb CPU	unknown	0.1357	2
48_31	moderate	48	101100111000101011111111101010111011011000110110	quimb CPU	unknown	0.4209	2
48_32	moderate	48	011101010111011110001110010101010101101110001110	quimb CPU	unknown	0.2656	2
56_33	moderate	56	11001001100101001111010100100010010101011111110111 01010000	quimb CPU	unknown	0.04102	2
64_34	moderate	64	0011010100010011001110101110100101001011001011 1001111011100110	quimb CPU	unknown	0.1631	2
40_35	hard	40	0101100110101111111000010101001110001001	quimb GPU	unknown	0.001953	3
48_36	hard	48	001111011011001100110011001000101001101010111000	quimb GPU	unknown	0.007812	3
48_37	hard	48	100101001010001101010100100101001001000101010001	quimb GPU	unknown	0.0009766	3
56_38	hard	56	10011001011110000000010011010101100001001001110 1 10110001	RCM CPU	unknown	0.001953	1
56_39	hard	56	10010111001101010101111111010110110110110000101 0 01010000	quimb GPU	unknown	0.001953	3
64_40	hard	64					0
64_41	hard	64	0001000111010011100110010011110111100000010001101 0001101010110010	Aer MPS	unstable_top1_vs_aggregate	0.167	1
48_42	very hard	48					0
56_43	very hard	56					0
64_44	very hard	64					0
72_45	very hard	72					0
80_46	very hard	80					0
88_47	very hard	88					0
96_48	very hard	96					0
104_49	very hard	104					0

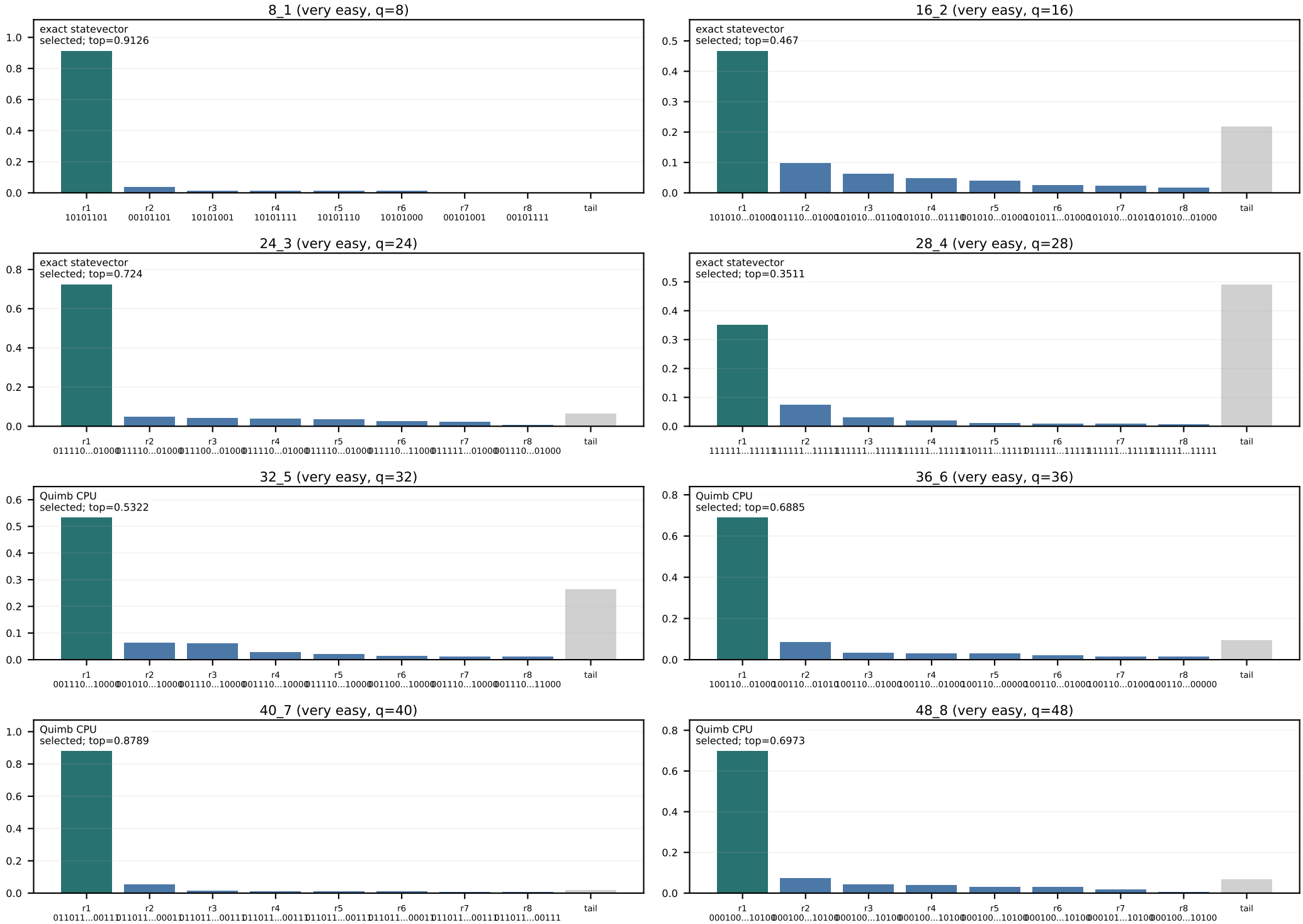
Conclusions

The practical output of the session is a 40-row candidate answer set and a clear split between reliable and weak evidence. The exact and high-top-fraction Quimb rows are the most defensible. The hard weak rows should be treated as provisional. The best next direction is not another unconstrained graph-MPS sweep: it is bounded MPO-unswapping or optimized-U3 GPU runs with calibration checks and immediate collector refreshes.

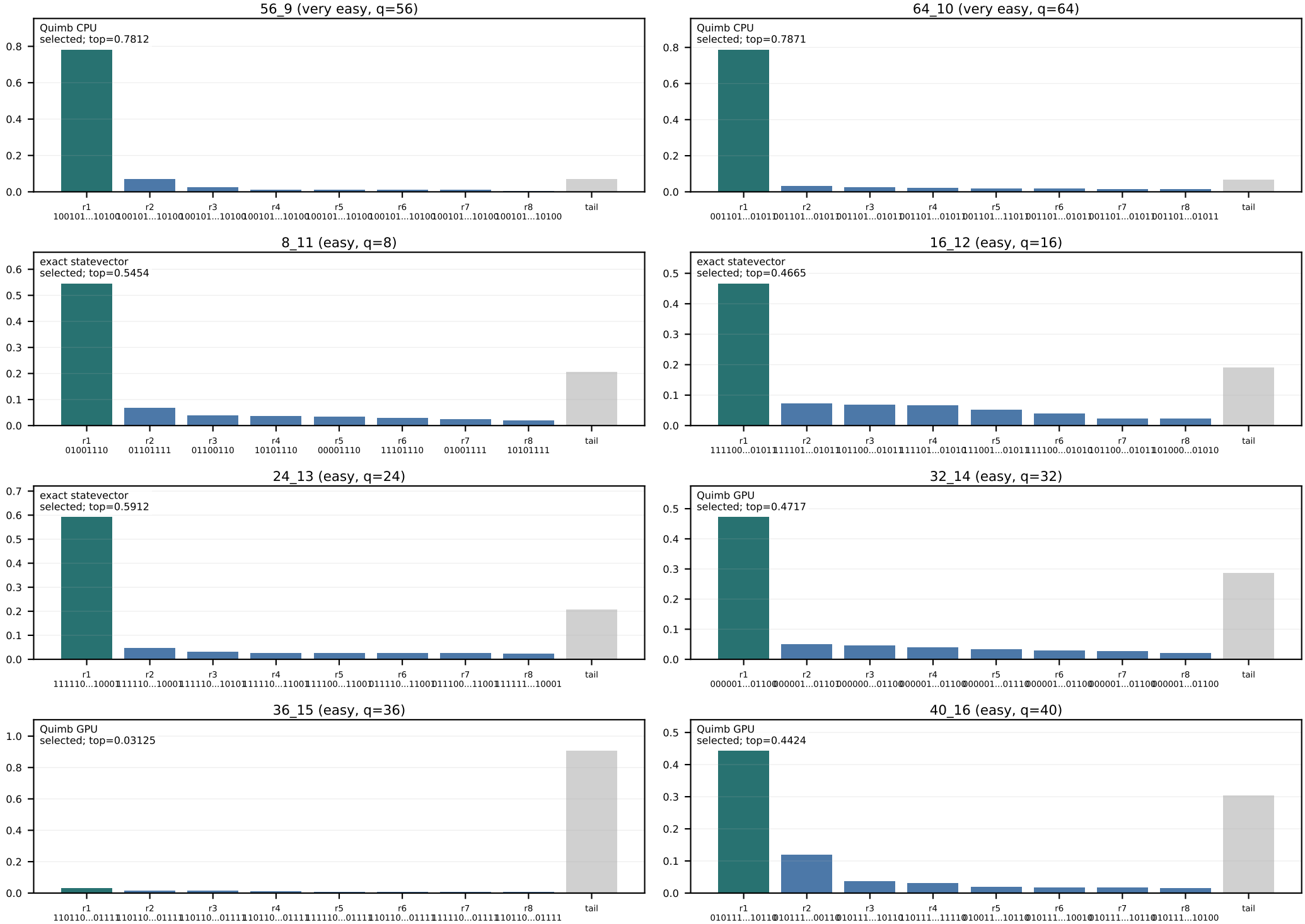
Appendix: Per-Problem Result Distributions

Each panel below shows the best distribution evidence retained for one challenge. Exact panels show the largest exact probabilities from statevector simulation. Sampled panels show the top observed bitstrings from Quimb MPS, MPO-unswap, or MPS distillation evidence; the grey tail is all unplotted probability or sample mass. Green bars mark a plotted bitstring that equals the selected candidate. Orange panels mean no answer was selected even though a low-confidence distribution was available. Bitstrings in plots are shortened for readability; the full selected bitstrings remain in the selected candidate table above. All bitstrings use Qiskit count order.

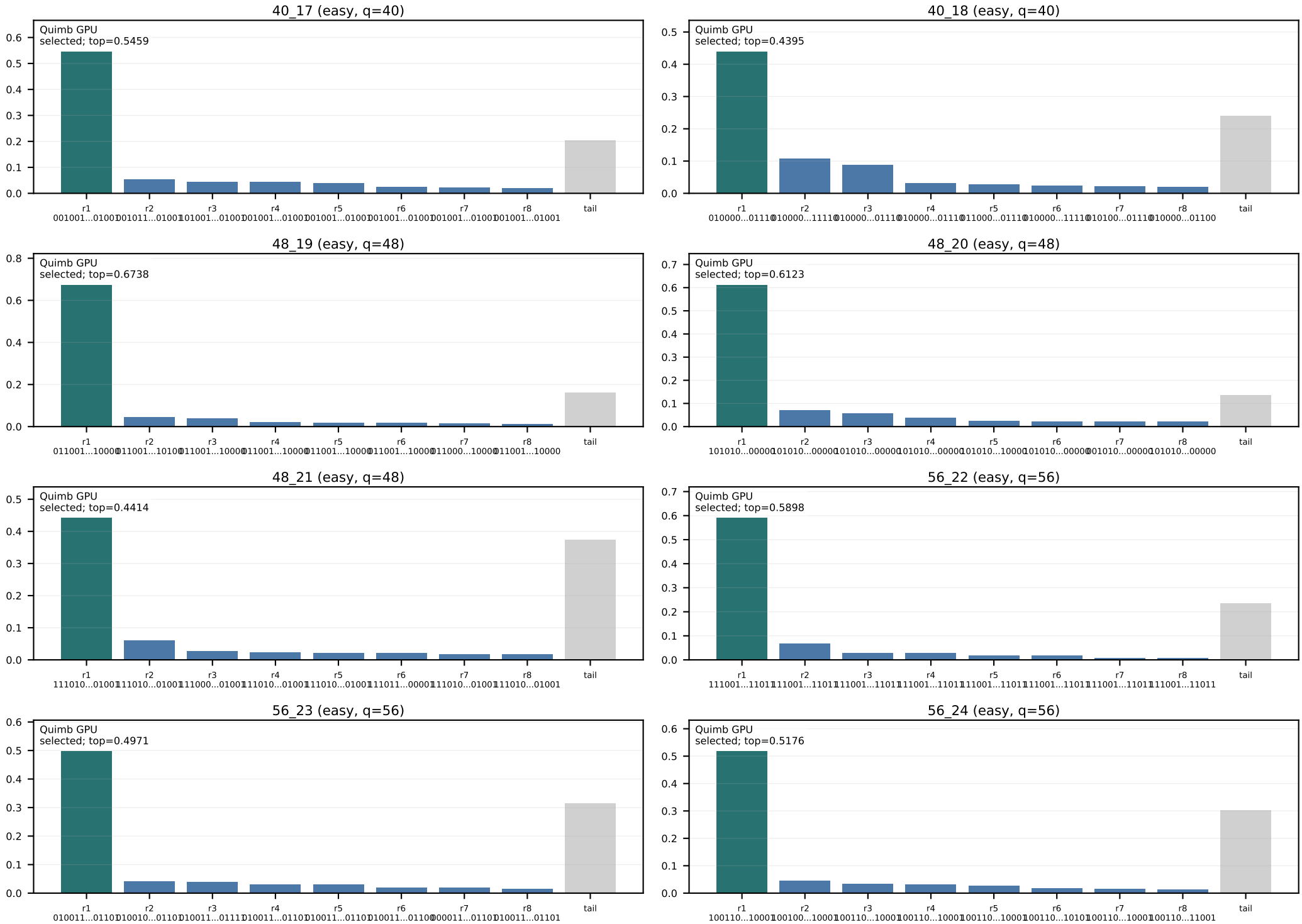
Per-problem result distributions, page 1/7



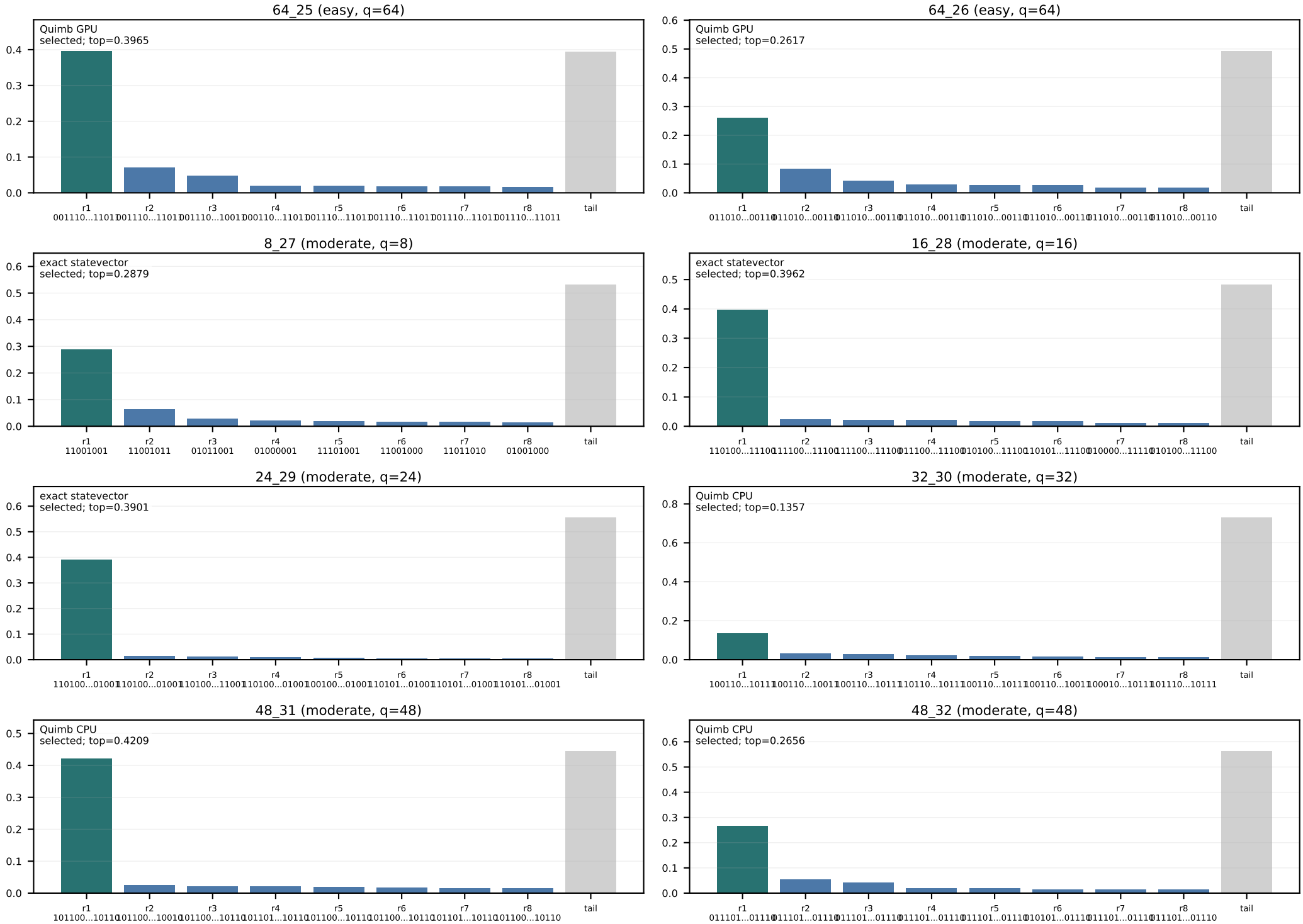
Per-problem result distributions, page 2/7



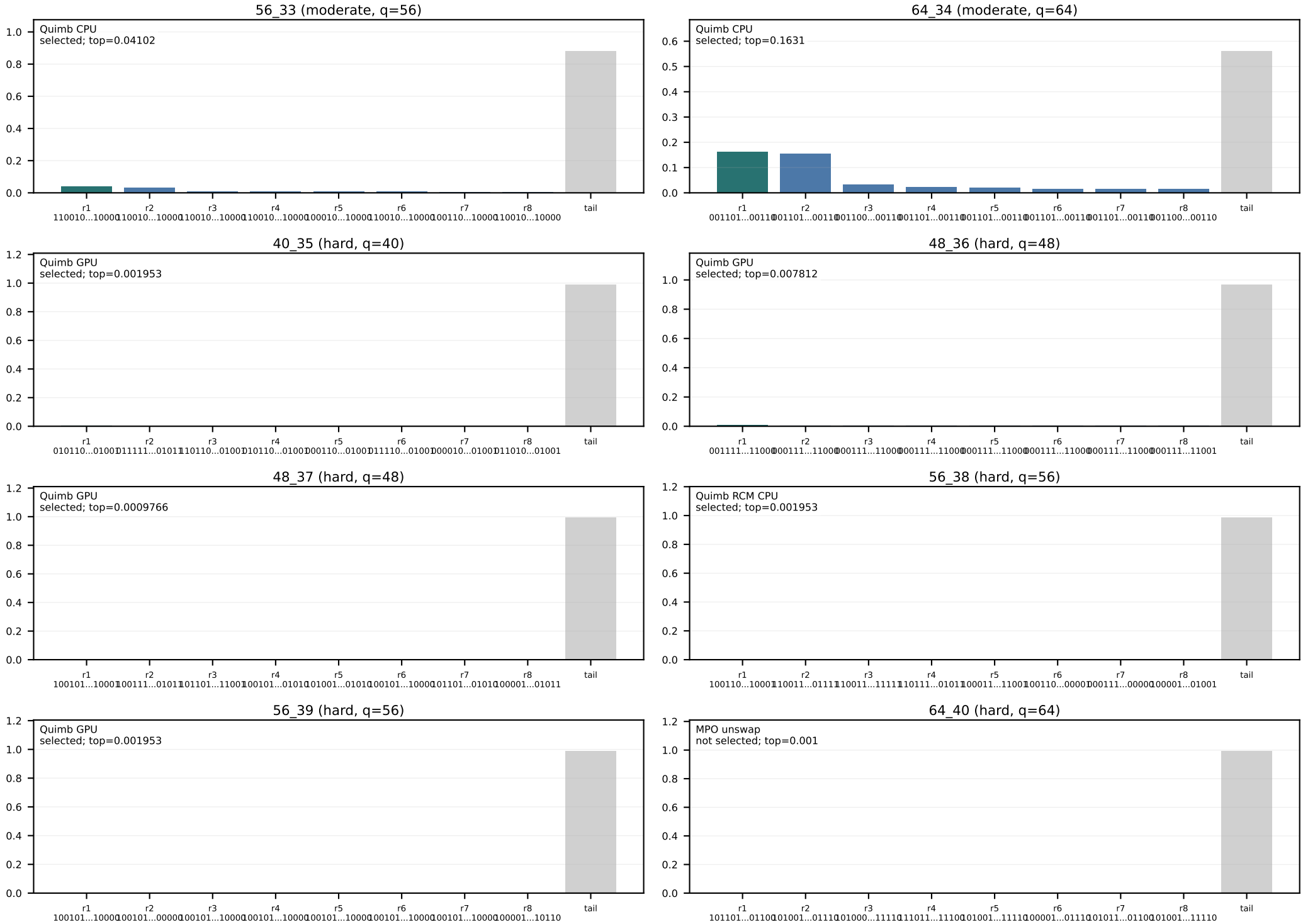
Per-problem result distributions, page 3/7

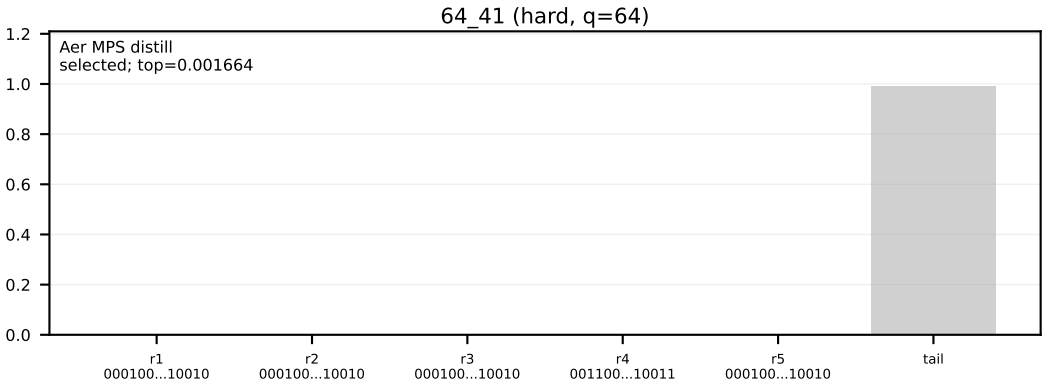


Per-problem result distributions, page 4/7



Per-problem result distributions, page 5/7





48_42 (very_hard, q=48)

no completed distribution

56_43 (very_hard, q=56)

no completed distribution

64_44 (very_hard, q=64)

no completed distribution

72_45 (very_hard, q=72)

no completed distribution

80_46 (very_hard, q=80)

no completed distribution

88_47 (very_hard, q=88)

no completed distribution

96_48 (very_hard, q=96)

no completed distribution

104_49 (very_hard, q=104)

no completed distribution

Distribution Metadata

The table records the source used in each appendix panel and the top plotted mass. For exact rows this is an exact probability; for sampled rows it is an observed sample fraction or aggregate support fraction. Rows with no completed distribution are left blank.

Challenge	difficulty	q	selected source	plotted source	top mass	appendix status
8_1	very easy	8	exact statevector	exact statevector	0.9126	selected
16_2	very easy	16	exact statevector	exact statevector	0.467	selected
24_3	very easy	24	exact statevector	exact statevector	0.724	selected
28_4	very easy	28	exact statevector	exact statevector	0.3511	selected
32_5	very easy	32	Quimb CPU	Quimb CPU	0.5322	selected
36_6	very easy	36	Quimb CPU	Quimb CPU	0.6885	selected
40_7	very easy	40	Quimb CPU	Quimb CPU	0.8789	selected
48_8	very easy	48	Quimb CPU	Quimb CPU	0.6973	selected
56_9	very easy	56	Quimb CPU	Quimb CPU	0.7812	selected
64_10	very easy	64	Quimb CPU	Quimb CPU	0.7871	selected
8_11	easy	8	exact statevector	exact statevector	0.5454	selected
16_12	easy	16	exact statevector	exact statevector	0.4665	selected
24_13	easy	24	exact statevector	exact statevector	0.5912	selected
32_14	easy	32	Quimb GPU	Quimb GPU	0.4717	selected
36_15	easy	36	Quimb GPU	Quimb GPU	0.03125	selected
40_16	easy	40	Quimb GPU	Quimb GPU	0.4424	selected
40_17	easy	40	Quimb GPU	Quimb GPU	0.5459	selected
40_18	easy	40	Quimb GPU	Quimb GPU	0.4395	selected
48_19	easy	48	Quimb GPU	Quimb GPU	0.6738	selected
48_20	easy	48	Quimb GPU	Quimb GPU	0.6123	selected
48_21	easy	48	Quimb GPU	Quimb GPU	0.4414	selected
56_22	easy	56	Quimb GPU	Quimb GPU	0.5898	selected
56_23	easy	56	Quimb GPU	Quimb GPU	0.4971	selected
56_24	easy	56	Quimb GPU	Quimb GPU	0.5176	selected
64_25	easy	64	Quimb GPU	Quimb GPU	0.3965	selected
64_26	easy	64	Quimb GPU	Quimb GPU	0.2617	selected
8_27	moderate	8	exact statevector	exact statevector	0.2879	selected
16_28	moderate	16	exact statevector	exact statevector	0.3962	selected
24_29	moderate	24	exact statevector	exact statevector	0.3901	selected
32_30	moderate	32	Quimb CPU	Quimb CPU	0.1357	selected
48_31	moderate	48	Quimb CPU	Quimb CPU	0.4209	selected
48_32	moderate	48	Quimb CPU	Quimb CPU	0.2656	selected
56_33	moderate	56	Quimb CPU	Quimb CPU	0.04102	selected
64_34	moderate	64	Quimb CPU	Quimb CPU	0.1631	selected
40_35	hard	40	Quimb GPU	Quimb GPU	0.001953	selected
48_36	hard	48	Quimb GPU	Quimb GPU	0.007812	selected
48_37	hard	48	Quimb GPU	Quimb GPU	0.0009766	selected
56_38	hard	56	Quimb RCM CPU	Quimb RCM CPU	0.001953	selected
56_39	hard	56	Quimb GPU	Quimb GPU	0.001953	selected
64_40	hard	64		MPO unswap	0.001	not selected
64_41	hard	64	Aer MPS distill	Aer MPS distill	0.001664	selected
48_42	very_hard	48				no completed distribution
56_43	very_hard	56				no completed distribution
64_44	very_hard	64				no completed distribution
72_45	very_hard	72				no completed distribution
80_46	very_hard	80				no completed distribution
88_47	very_hard	88				no completed distribution
96_48	very_hard	96				no completed distribution
104_49	very_hard	104				no completed distribution